

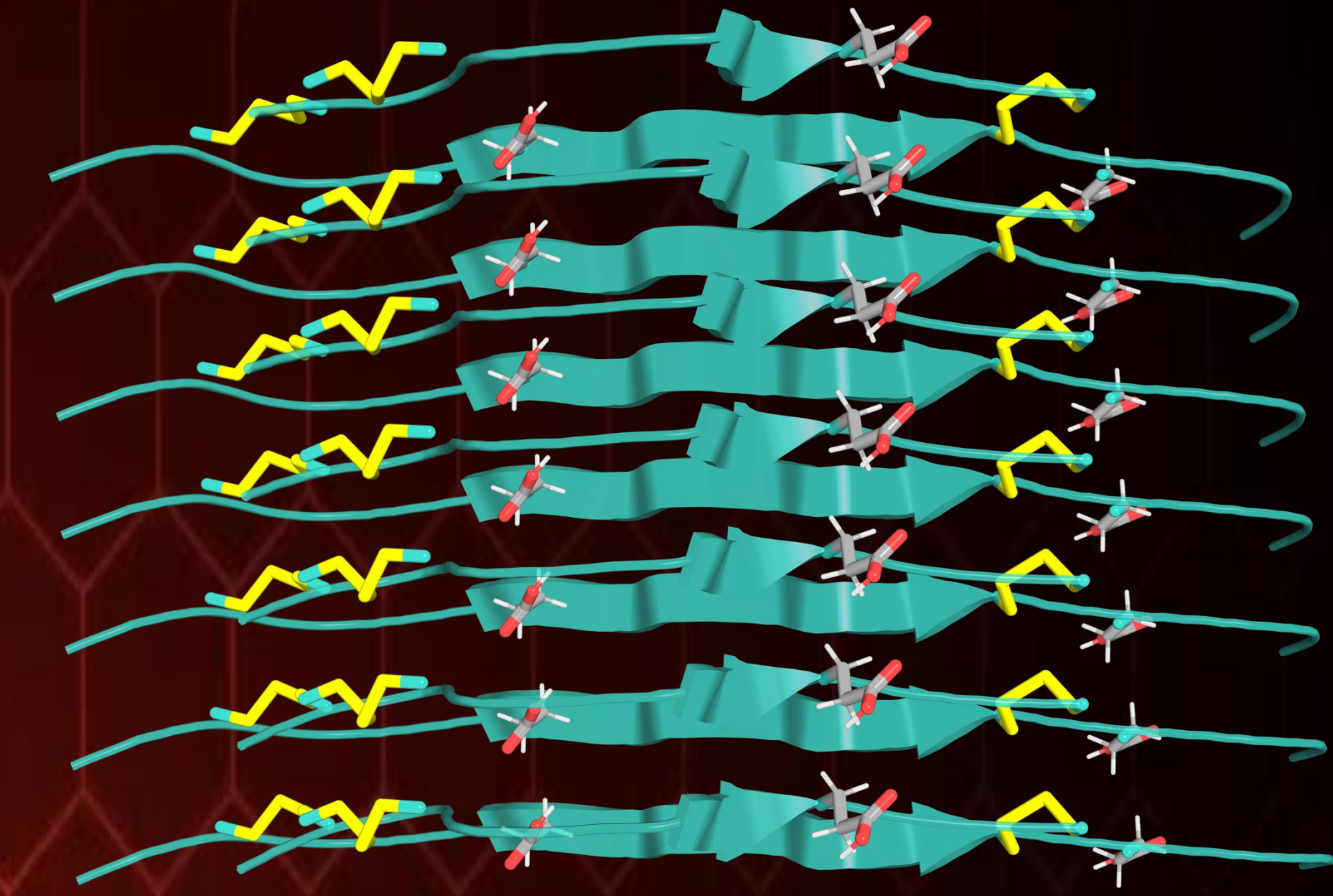
pH-Dependent Free Energies of Insulin Fibrillation from Protonation Ensemble Statistics

Stefan Hervø-Hansen, Kento Kasahara, Nobuyuki Matubayasi



1. Insulin Fibrillation and Protein Electrostatics

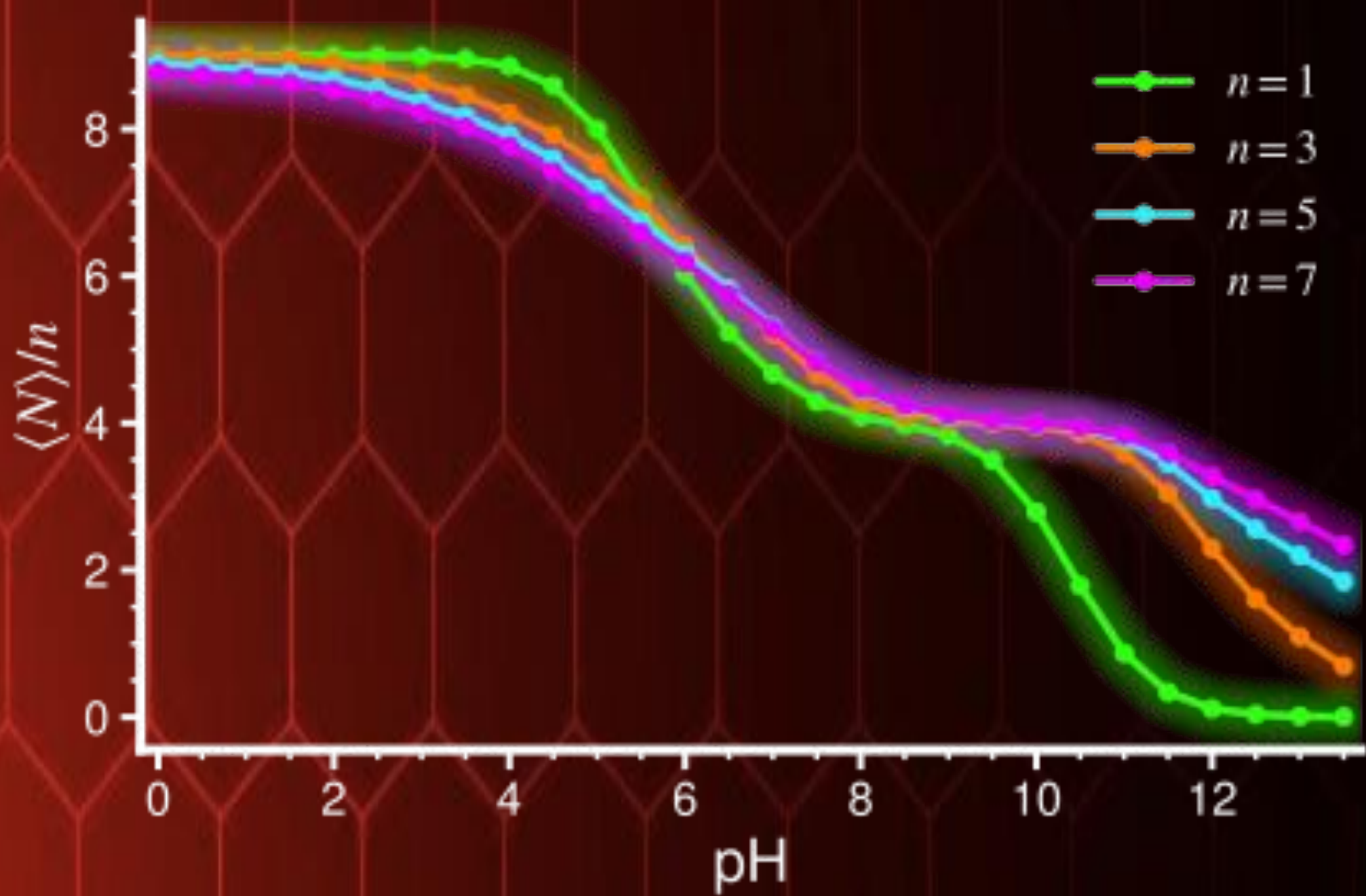
At low and near neutral pH insulin forms amyloid fibrils



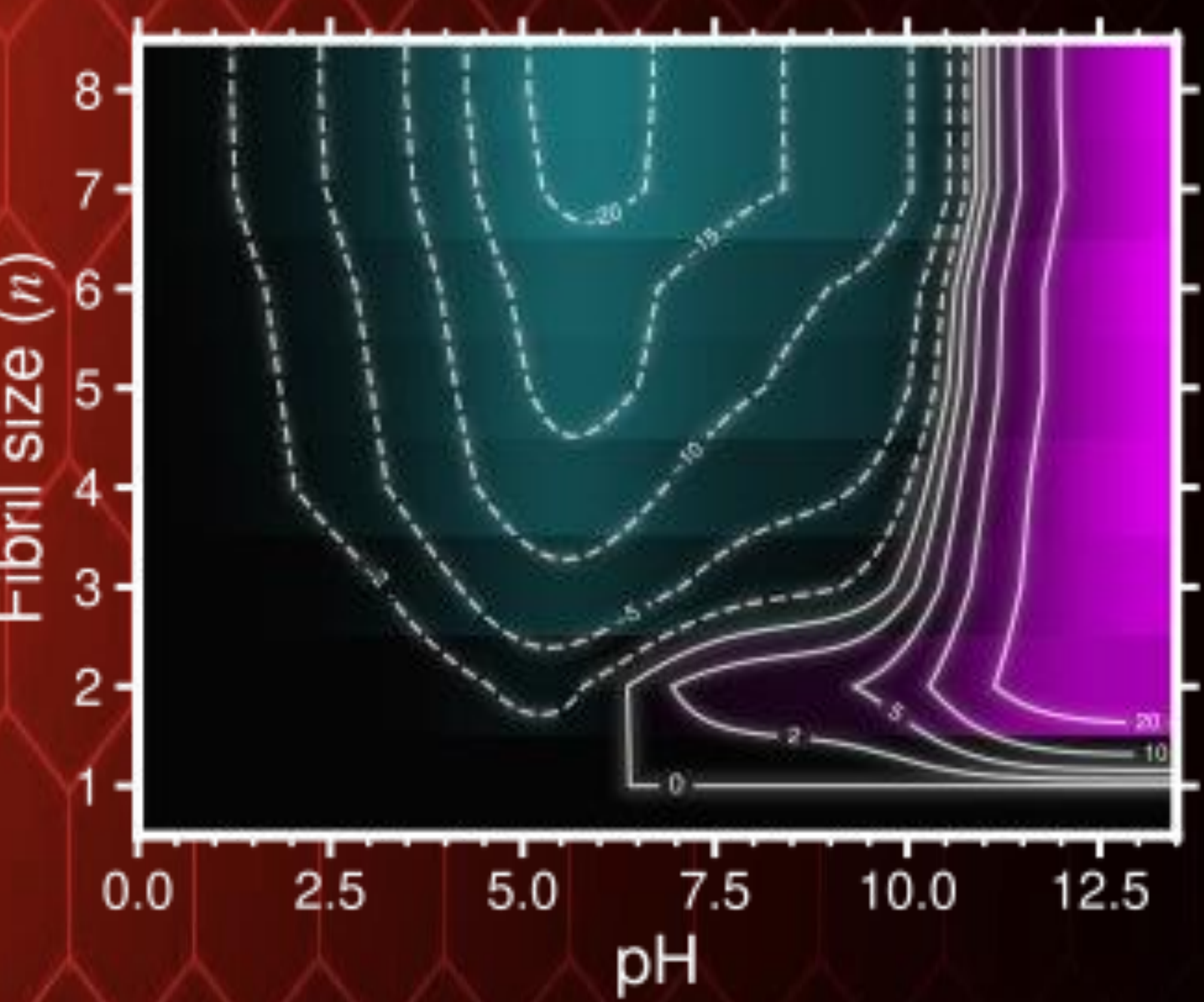
Pharmacologically inactive and potentially cytotoxic

4. Site-Unresolved Titration Curves of Insulin Fibrils

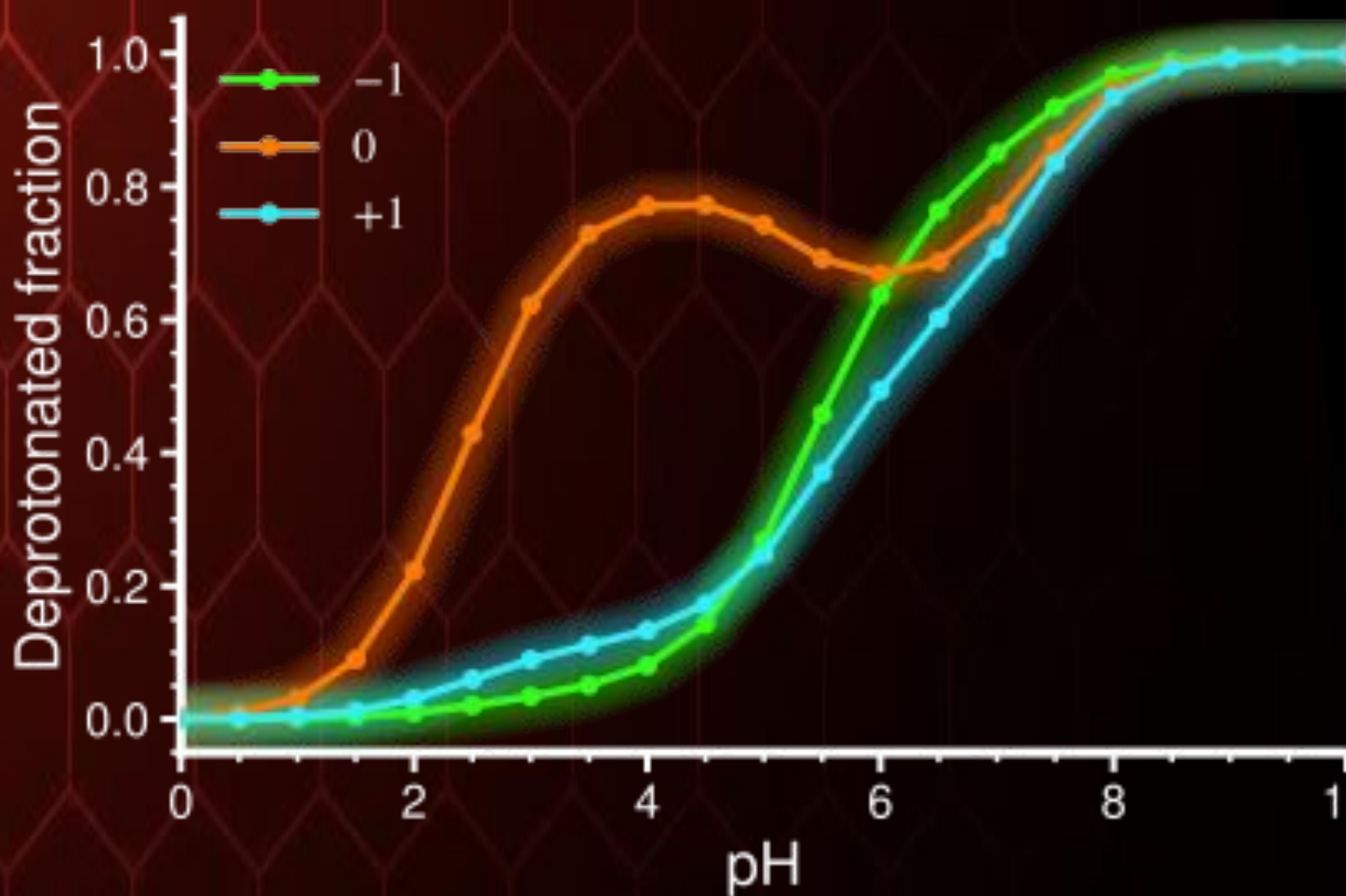
The site-unresolved constitutes the minimum necessary information to obtain pH-dependent free energies



5. Free Energy Landscape and the Problem of Mechanism



6. Insights from Site-resolved Titration Curves and Structural Analysis



Multistep titration, including, *re-protonation*, caused by strongly correlated Glu residues, forming unique interaction patterns

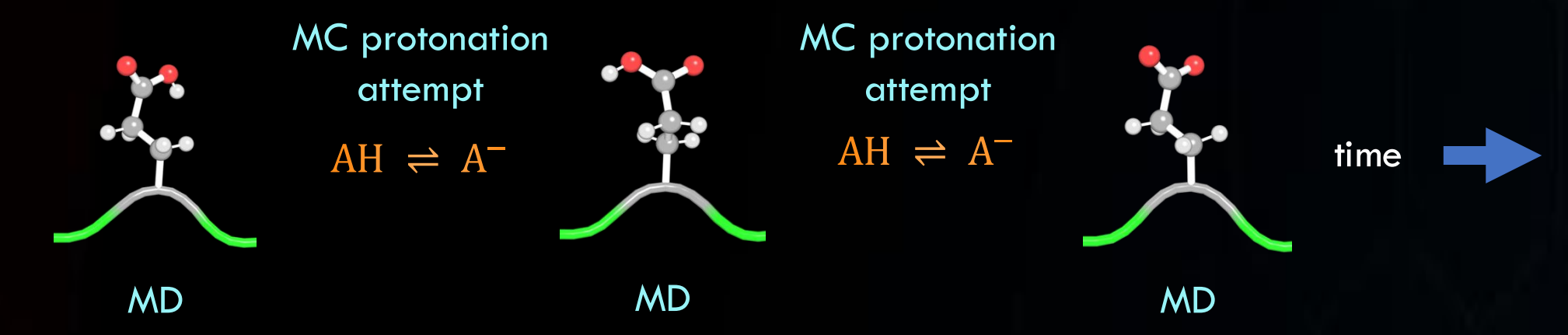
2. pH-Replica Exchange Molecular Dynamics

Probability distribution given by

$$P(\psi, \mathbf{X}, \mathbf{n}) \propto e^{-\beta U(\psi, \mathbf{X}, \mathbf{n})} e^{\beta \mu_{\text{H}^+} N(\mathbf{n})}$$

$(\psi, \mathbf{X})$: Protein and solvent conformations sampled by molecular dynamics

$\mathbf{n}$: Protonation state vector sampled by Metropolis Monte Carlo



The conformational sampling is conducted in the presence of explicit water, while the protonation state vector is sampled in the presence of implicit water

3. PRIME: Protonation-state Reweighting In Multi-pH Ensembles

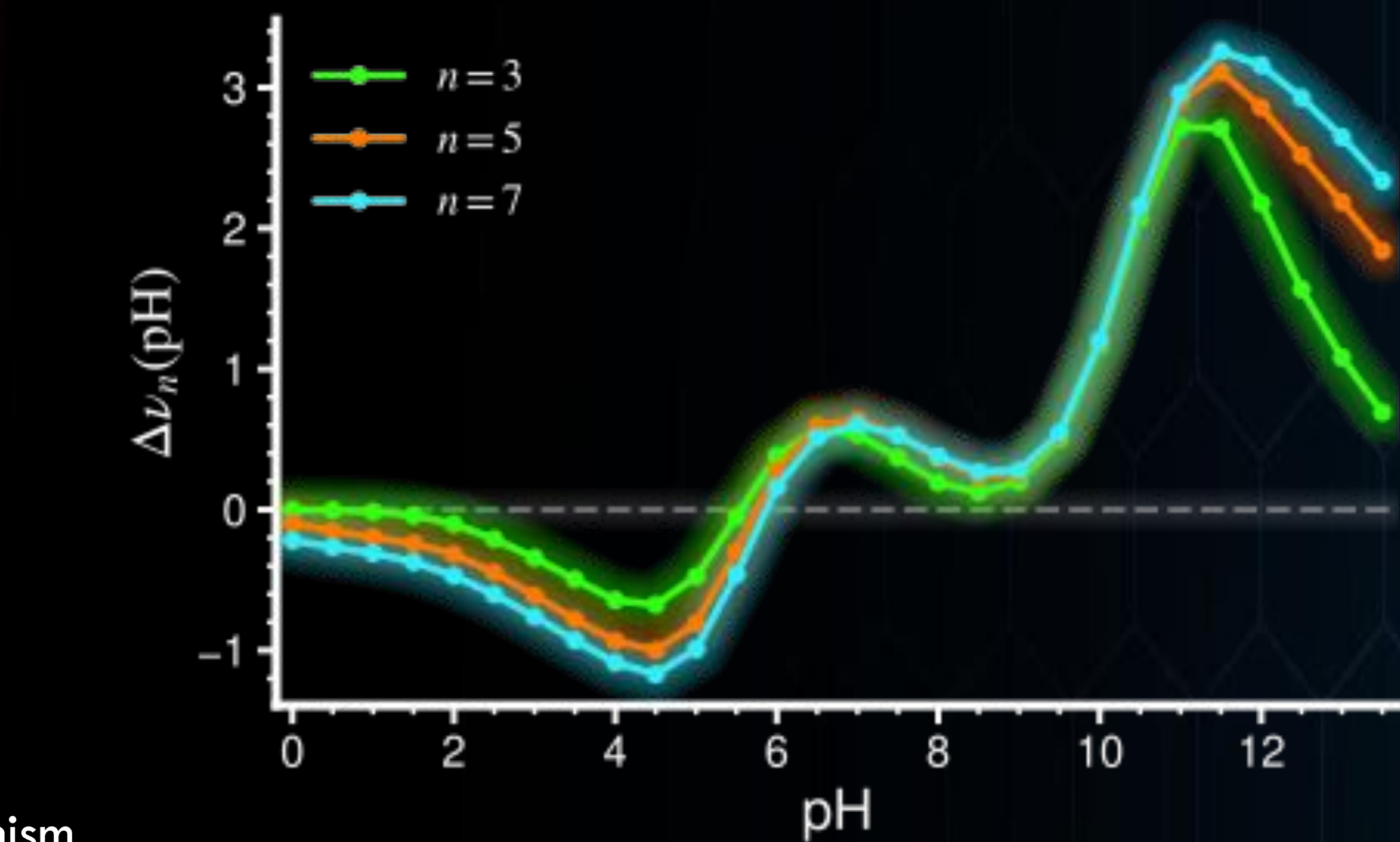
We utilize a minimal variance free energy estimator between pH state

$$\sum_{N=0}^{N_{\text{max}}} \frac{H_{\text{tot}}(N) \exp[-\beta \Upsilon_m + \beta \mu_{\text{H}^+}^m N]}{\sum_{j=1}^K M_j \exp[-\beta \Upsilon_j + \beta \mu_{\text{H}^+}^j N]} = 1 \quad \forall m = 1, \dots, K$$

Equivalent to MBAR but uses total protonation values, thus being naturally discrete and reassembles WHAM

Force Fields:
Protein: Amber's ff10 + carboxylate modifications
Explicit Water: TIP3P
Implicit Water: Onufriev, Bashford, Case model, *Proteins*, 2004

We define a differential response to quantify the effect of insulin fibrillation



$$\Delta \nu_n(\text{pH}) = \frac{\langle N \rangle_n}{n} - \langle N \rangle_{n=1}$$

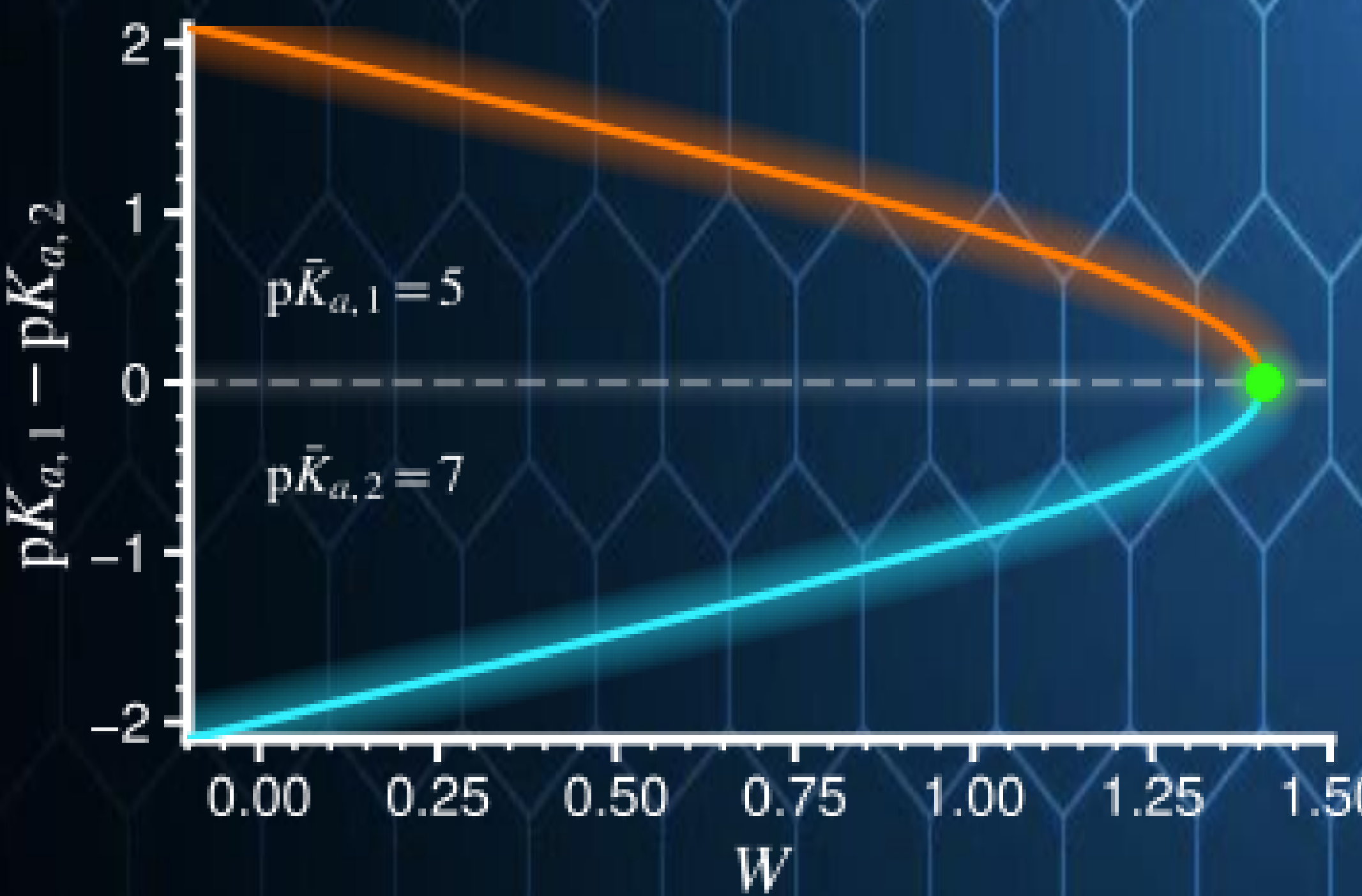
$\Delta \nu_n(\text{pH}) < 0$: Fibril formation increases proton affinity

$\Delta \nu_n(\text{pH}) > 0$: Fibril formation decreases proton affinity

At pH = 4.5 fibrillation causes proton retention

Due to this retention of protons, the process of fibril formation becomes thermodynamically unfavorable at high pH relative to the well-protonated conditions at low pH

BUT
Thermodynamics lives in *macroscopic* space
 $\underbrace{(\{pK_{a,1}, pK_{a,2}\})}_{\text{macroscopic}} \rightarrow \underbrace{(\{pK_{a,1}, pK_{a,2}, W\})}_{\text{microscopic}}$
 $0 \leq W \leq W_{\text{max}}$
Mechanism lives in *microscopic* space



For a diprotic acid, this transformation yields a continuous solution manifold constrained by two physical extremes: the maximum coupling energy and non-interaction limit(s)

Low pH:
No electrostatic frustration → Compact interface

Intermediate pH:
Mixed protonation → Glu-Glu electrostatic network

High pH:
Charge accumulation → Inter-monomer repulsion

Conclusion

The pH-dependent free energy landscape of insulin fibrillation reveals a strong driving force for fibrillation at weakly acidic conditions, facilitated by proton retention. Structural analysis and site-resolved titration curves confirms this thermodynamic preference is governed by a correlated Glu-Glu network that balances charge accumulation to maximize inter-monomer contacts